

Computer Networks

Network Layer — Complete Notes

Routing Algorithms: Shortest Path • Flow-Based
Distance Vector • Broadcast • Multicast
IP Addressing: IPv4 & IPv6

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Subject: Computer Networks

Topic: Network Layer & IP Addressing

Prepared with Claude AI • Exam Revision Notes

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1. Introduction to the Network Layer

The **Network Layer** is Layer 3 of the OSI model. Its main job is to get data packets from the **source host** to the **destination host**, even if they are on completely different networks separated by many routers.

Real-world analogy: Think of the Network Layer as the Post Office of the internet. You give it a letter (packet) with an address, and it figures out the best route to deliver it — passing through multiple sorting centers (routers) on the way.

Key Responsibilities

- Logical Addressing — assigns IP addresses to uniquely identify every device
- Routing — chooses the best path through the network for each packet
- Packet Forwarding — moves each packet from one router to the next (hop-by-hop)
- Fragmentation & Reassembly — breaks large packets into smaller ones if needed
- Error Handling & Congestion Control — handles errors and avoids network overload

Key Device: The Router

- Routers operate at the Network Layer
- They read the destination IP address in each packet
- They look up their routing table to decide the next hop
- They forward the packet out of the correct interface
- This process repeats at every router until the packet reaches its destination

2. Shortest Path Algorithm (Dijkstra's)

The Shortest Path Algorithm finds the path with the **minimum total cost** between a source and a destination node. It is the most widely used routing algorithm, forming the foundation of the **OSPF (Open Shortest Path First)** protocol.

Real-world analogy: Just like Google Maps checks all possible routes and picks the shortest/fastest one, Dijkstra's algorithm checks all possible paths in the network and picks the one with minimum total cost.

How It Works — Step by Step

- Step 1:** Assign cost 0 to the source node, and ∞ (infinity) to all other nodes.
- Step 2:** Mark all nodes as UNVISITED.
- Step 3:** From the current node, look at all unvisited neighbors.
- Step 4:** Calculate tentative cost: $\text{tentative} = \text{dist}[\text{current}] + \text{weight}(\text{current}, \text{neighbor})$
- Step 5:** If tentative cost < current known cost for that neighbor → UPDATE it.
- Step 6:** Mark the current node as VISITED (we are done with it).
- Step 7:** Move to the unvisited node with the smallest known cost.
- Step 8:** Repeat steps 3–7 until the destination node is visited.

Key Formula: $\text{dist}[v] = \text{dist}[u] + \text{weight}(u, v)$ (if this is less than current $\text{dist}[v]$)

Worked Example

Network: A–B (cost 4), A–C (cost 2), B–D (cost 3), C–D (cost 5), D–E (cost 1). Find shortest path from A to E.

Step	Visited Node	dist[A]	dist[B]	dist[C]	dist[D]	dist[E]
Start	—	0	∞	∞	∞	∞
1	A	0	4	2	∞	∞
2	C	0	4	2	7	∞
3	B	0	4	2	7	∞
4	D	0	4	2	7	8
5	E ✓	0	4	2	7	8

Shortest path: $A \rightarrow C \rightarrow D \rightarrow E$, Total cost = $2 + 5 + 1 = 8$

Advantages

- Always finds the optimal (shortest) path
- Works well with any link costs
- Used in real-world OSPF protocol
- Guaranteed to converge

Disadvantages

- Needs complete knowledge of network topology
- Expensive for very large networks
- Every router must store the full graph
- $O(V^2)$ time complexity (basic version)

Key Facts

- Time Complexity: $O(V^2)$ with simple array, $O(E \log V)$ with priority queue
- Type: Greedy algorithm (always picks minimum cost next)
- Used in: OSPF (Open Shortest Path First) — the most popular link-state protocol
- Requires: Global knowledge of the entire network topology

3. Flow-Based Routing

Flow-Based Routing chooses paths based on **current traffic load** on each link, not just the physical distance or cost. It avoids congested links even if they are physically shorter, to minimize overall delay.

Real-world analogy: Imagine two roads from home to college. Road A is 2 km but completely jammed. Road B is 5 km but completely free. You'll take Road B — that's exactly what flow-based routing does!

The Key Delay Formula

$$\text{Delay on a link} = \text{Flow} / (\text{Capacity} - \text{Flow})$$

As flow approaches capacity, the denominator (Capacity – Flow) gets very small, making delay shoot up to infinity. So the algorithm avoids heavily loaded links.

Capacity	Current Flow	Formula	Delay
100 Mbps	20 Mbps	$20 / (100 - 20) = 20/80$	0.25 (Low ✓)
100 Mbps	50 Mbps	$50 / (100 - 50) = 50/50$	1.0 (Medium)
100 Mbps	90 Mbps	$90 / (100 - 90) = 90/10$	9.0 (High ✗)
100 Mbps	99 Mbps	$99 / (100 - 99) = 99/1$	99.0 (Very High ✗✗)

Static Flow-Based

- Routes calculated once at start
- Based on predicted/expected traffic
- Simple but doesn't adapt to changes
- Can become outdated quickly

Dynamic Flow-Based

- Routes recalculated periodically
- Based on real-time traffic measurements
- Adapts to changing network conditions
- More complex but more accurate

Key Facts

- Main goal: Minimize delay by avoiding congested links
- Requires: Real-time or predicted traffic data for each link
- Advantage: Prevents bottlenecks, balances load across the network
- Disadvantage: Needs global traffic knowledge — expensive to collect and update

4. Distance Vector Routing (Bellman-Ford)

In Distance Vector Routing, each router maintains a **distance vector table** — a list of all destinations and the cost to reach each one. Routers periodically share this table with their **directly connected neighbors**. Based on what neighbors report, each router updates its own table.

Real-world analogy: You're new to a city and want to reach a hospital. You ask your neighbor — he says it's 3 km from him. You know he's 2 km from you. So you calculate: hospital is $2 + 3 = 5$ km from you. That's distance vector routing!

The Bellman-Ford Equation

$$D(x, y) = \min \text{ over all neighbors } v \{ \text{cost}(x, v) + D(v, y) \}$$

This means: the distance from x to y is the minimum of (cost to reach any neighbor v) + (that neighbor's distance to y).

Step-by-Step Process

- Each router starts knowing only its directly connected neighbors and their costs.
- Periodically (e.g., every 30 seconds in RIP), each router sends its distance table to all neighbors.
- When a router receives a neighbor's table, it checks: can I reach any destination cheaper via this neighbor?
- If yes → update the routing table entry with the new cost and mark this neighbor as the next hop.
- This process repeats until no more updates occur → the network has converged.

Example Distance Vector Tables (3 Routers: A — B — C)

Link costs: $A \leftrightarrow B = 2$, $B \leftrightarrow C = 3$

Router A		
Dest	Cost	Via
A	0	self
B	2	B
C	5	B

Router B		
Dest	Cost	Via
A	2	A
B	0	self
C	3	C

Router C		
Dest	Cost	Via
A	5	B
B	3	B
C	0	self

■ The Count-to-Infinity Problem

- Scenario: A–B–C network. A can reach C via B in 5 hops.
- Now the A–B link BREAKS. B should know C is unreachable from A.
- But B still thinks A can reach C! So B tells A 'C is 6 hops via me'.
- A believes B, updates to 7. B hears this, updates to 8. Keeps going up...
- This loop continues until the cost reaches infinity (e.g., 16 in RIP = unreachable).
- Fix: Split Horizon (don't advertise a route back to where you learned it from).

Key Facts

- Algorithm: Bellman-Ford (distributed version)
- Used in: RIP — Routing Information Protocol (older internet routing)
- Update method: Each router shares full table with neighbors periodically
- Problem: Count-to-Infinity — bad news spreads very slowly
- Fix: Split Horizon, Poison Reverse, Route Poisoning
- Comparison: Slower convergence than Link State (Dijkstra), but uses less memory

5. Broadcast Routing

Broadcast Routing sends a packet from **one source to ALL nodes** in the entire network. Every router and device must receive a copy of the packet.

Real-world analogy: A school principal picks up the mic and makes an announcement — every student in every classroom hears it simultaneously. That's broadcast!

Method 1: Flooding (Simple but Wasteful)

- Router receives a packet → sends it out on ALL outgoing links (except the one it came from).
- This is simple to implement — no routing table needed.
- Problem: The same packet gets duplicated and circulates forever, causing broadcast storms.
- Fix: Use sequence numbers — routers discard packets they've already seen.

Method 2: Spanning Tree Broadcasting (Efficient ✓)

- Build a Spanning Tree — a tree structure that covers ALL nodes with NO cycles/loops.
- The broadcast packet travels ONLY along the tree edges.
- Each node receives the packet exactly once — no duplicates, no loops!
- This is the standard method used in practice.

Method 3: Reverse Path Forwarding (RPF)

- When a packet arrives at a router, check: did it arrive on the link that is the shortest path BACK to the source?
- If YES → forward the packet out on all other links.
- If NO → discard the packet (it's a duplicate coming via a non-optimal path).
- Smart, loop-free, and doesn't require building an explicit spanning tree.

Method	How it Works	Duplicates?	Loops?	Used?
Flooding	Forward on all links	Many	Yes ✗	Rarely
Spanning Tree	Forward only on tree edges	None	No ✓	Yes ✓
Reverse Path (RPF)	Check if best path to source	Minimal	No ✓	Yes ✓

Key Facts

- Definition: One source → ALL nodes in the network
- Best method: Spanning Tree (no duplicates, no loops)
- RPF is a clever trick — doesn't need explicit tree construction
- Used in: ARP (Address Resolution Protocol), OSPF hello packets, network discovery
- Difference from Multicast: Broadcast goes to ALL; Multicast goes to a specific GROUP

6. Multicast Routing

Multicast Routing sends a packet from **one source to a specific GROUP of nodes** — not all nodes (broadcast) and not just one node (unicast). Only members who have explicitly joined the group receive the packet.

Real-world analogy: A WhatsApp group! When you send a message in your 'DSA Study Group', only the 5 members of that group receive it — not your entire contact list. That's multicast.

Unicast vs Broadcast vs Multicast

Type	Destination	Example
Unicast	1 specific node	One-to-one call / WhatsApp message to 1 friend
Broadcast	ALL nodes	School announcement over intercom
Multicast ✓	Specific GROUP of nodes	WhatsApp group / Zoom meeting / IPTV channel

How Multicast Routing Works

- Nodes that want to receive multicast traffic JOIN a multicast group (e.g., IP group 224.0.0.1).
- A multicast spanning tree is built — but ONLY branches that lead to group members are kept.
- Branches leading to non-members are PRUNED (cut off).
- The packet travels only along the tree branches that reach group members.
- Non-members don't receive the packet at all — saves bandwidth!

Source-Based Tree

- Separate tree built per source
- Optimal path for each source
- More memory/computation needed
- Better for few sources

Shared Tree

- One shared tree for whole group
- Uses a common Rendezvous Point
- Less memory, more scalable
- Path may not be optimal

Key Facts & Real-World Uses

- Definition: One source → specific GROUP of nodes (group members only)
- IP Range for Multicast groups: 224.0.0.0 to 239.255.255.255 (Class D addresses)
- Protocol used: IGMP (Internet Group Management Protocol) — nodes join/leave groups
- Used in: Video conferencing (Zoom/Teams), IPTV, online gaming, stock market data feeds
- Key benefit: Saves bandwidth — packet sent once on shared links, not separately to each member

7. Quick Comparison of All Routing Algorithms

Algorithm	Basis	Info Shared	Protocol	Key Problem
Shortest Path (Dijkstra)	Link cost/weight	Full graph (global)	OSPF	Needs full topology
Flow-Based	Traffic load	Traffic stats	Load bal.	Needs real-time data
Distance Vector (Bellman-Ford)	Hop count/cost	Neighbor tables	RIP	Count to Infinity
Broadcast	All destinations	Spanning tree	ARP/OSPF	Flooding = wasteful
Multicast	Group members	Group membership	IGMP/PIM	Group management

8. IP Addressing — IPv4

Every device on the internet needs a unique address — just like your home needs a unique postal address. An **IP (Internet Protocol) address** is that unique identifier. **IPv4** is the original addressing scheme, using 32-bit addresses.

*Real-world analogy: An IPv4 address is like a 4-part locker combination. Each part goes from 0 to 255.
Example: 192.168.1.1*

IPv4 Address Structure

Format: **192 . 168 . 1 . 1**

192	168	1	1
1st Octet	2nd Octet	3rd Octet	4th Octet
8 bits	8 bits	8 bits	8 bits
11000000	10101000	00000001	00000001

Key Facts — IPv4

- Total bits: 32 bits (4 octets × 8 bits each)
- Each octet range: 0 to 255
- Total possible addresses: 2^{32} = approximately 4.3 Billion
- Format: Dotted Decimal Notation (e.g., 192.168.1.1)
- Problem: We ran out of IPv4 addresses around 2011!
- Workaround: NAT (Network Address Translation) lets multiple devices share one public IP

9. IPv4 Address Classes

IPv4 addresses are divided into **5 classes (A, B, C, D, E)** based on the first octet value. Think of them like different zones in a city — big colonies (Class A), medium societies (Class B), small apartments (Class C).

Class	1st Octet	Starts With	Network / Host Split	No. of Networks	Hosts per Network	Use
A	1–126	0...	Net . H . H . H	128	16 Million	Giant orgs
B	128–191	10...	Net . Net . H . H	16,384	65,534	Universities
C	192–223	110...	Net . Net . Net . H	2 Million	254	Home/SME
D	224–239	1110...	Reserved for Multicast	—	—	Multicast
E	240–255	1111...	Reserved for Research	—	—	Research

Special Reserved Addresses to Remember

- 127.0.0.1 (loopback / localhost) — refers to your own machine. Used to test network software.
- 0.0.0.0 — default route (means 'any network')
- 255.255.255.255 — limited broadcast (send to everyone on local network)
- 10.x.x.x, 172.16–31.x.x, 192.168.x.x — Private IP ranges (used inside LANs, not on internet)

10. IP Addressing — IPv6

IPv6 was created to solve the IPv4 address exhaustion problem. It uses **128-bit addresses** written in hexadecimal, giving an astronomically large number of possible addresses. IPv6 also introduced several improvements over IPv4.

Real-world analogy: If IPv4 is like old 6-digit PIN codes (running out!), IPv6 is like a new system with 32-character codes — will NEVER run out!

IPv6 Address Structure

Format: **2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370 : 7334**

2001	0db8	85a3	0000	0000	8a2e	0370	7334
Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8
16 bits	16 bits	16 bits	16 bits	16 bits	16 bits	16 bits	16 bits

Shortening Rules — How to Write IPv6 More Easily

Rule 1: Remove leading zeros from each group.

- 0db8 → db8, 0000 → 0, 0370 → 370

Rule 2: Replace one or more consecutive groups of all-zeros with :: (double colon). Can only be used ONCE per address!

- 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- → 2001:db8:85a3::8a2e:370:7334 (after removing leading zeros and replacing 0000:0000 with ::)

New Features in IPv6

- No Broadcast: IPv6 eliminated broadcast. Uses multicast and anycast instead — more efficient.
- Built-in Security (IPSec): Encryption and authentication are mandatory in IPv6, not optional.
- SLAAC (Stateless Address Auto-Configuration): Devices can assign themselves an IPv6 address automatically — no DHCP server needed!
- Fixed 40-byte header: Simpler and faster for routers to process than IPv4's variable header.
- Better QoS: Flow label field lets routers identify and prioritize specific traffic flows.
- No NAT needed: Every device can have its own globally unique IPv6 address.

Key Facts — IPv6

- Total bits: 128 bits (8 groups × 16 bits each)
- Format: Hexadecimal with colons (e.g., 2001:db8::1)
- Total addresses: $2^{128} = 340$ undecillion (340,000,000,000,000,000,000,000,000,000,000)
- Multicast range: FF00::/8
- Loopback address: ::1 (equivalent of 127.0.0.1 in IPv4)
- Will NEVER run out — every grain of sand on Earth could have billions of IP addresses!

11. IPv4 vs IPv6 — Full Comparison

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Format	Decimal with dots (192.168.1.1)	Hex with colons (2001:db8::1)
Total Addresses	~4.3 Billion (2^{32})	340 Undecillion (2^{128})
Status	Exhausted since ~2011	Will never run out
Header Size	Variable (20–60 bytes), complex	Fixed 40 bytes, simpler
Security (IPSec)	Optional (add-on)	Mandatory (built-in)
Auto-Config	Needs DHCP server	SLAAC — self-configures!
Broadcast	Yes (can cause congestion)	No — uses Multicast instead
NAT Required	Yes (to share addresses)	No — every device has unique IP
Fragmentation	Done by routers and hosts	Done by sender only
QoS Support	Limited (Type of Service field)	Better (Flow Label field)
Loopback Address	127.0.0.1	::1
Example	192.168.1.1	2001:0db8:85a3::8a2e:370:7334

12. Quick Revision Summary

Read this page just before your exam! All the key facts in one place.

Network Layer — What to Remember

- Network Layer = Layer 3 of OSI = Routing packets from source to destination
- Key device: Router | Key function: Routing + Forwarding + Logical Addressing

Routing Algorithms — One-Line Summary

- Shortest Path (Dijkstra): Finds min-cost path. Uses global topology. → OSPF protocol
- Flow-Based: Avoids congested links. Delay = Flow/(Capacity-Flow). Needs real-time data.
- Distance Vector (Bellman-Ford): Routers share tables with neighbors. Slow. → RIP. Problem: Count-to-Infinity
- Broadcast: One → ALL. Use Spanning Tree to avoid loops. RPF is another smart method.
- Multicast: One → GROUP. Members join group. Non-member branches are pruned. → IGMP/IPTV

IPv4 — What to Remember

- 32-bit address, 4 octets, dotted decimal: 192.168.1.1
- Total: $2^{32} = \sim 4.3$ Billion addresses. Exhausted around 2011.
- Class A (1-126): 16 million hosts | Class B (128-191): 65534 hosts | Class C (192-223): 254 hosts
- Class D (224-239): Multicast | Class E (240-255): Research
- 127.x.x.x = Loopback (localhost) | Private ranges: 10.x, 172.16-31.x, 192.168.x

IPv6 — What to Remember

- 128-bit address, 8 groups of 4 hex digits, colon-separated: 2001:db8::1
- Total: $2^{128} = 340$ Undecillion. Will NEVER run out!
- Shortening: Remove leading zeros, replace consecutive 0000 groups with :: (once only)
- Loopback: ::1 | No broadcast | Built-in IPSec security | SLAAC auto-configuration
- Fixed 40-byte header = faster processing | No NAT needed

Exam Tip — Easy Memory Tricks

Topic	Memory Trick
Dijkstra	Greedy like GPS — always picks the cheapest next step
Flow-Based	Like avoiding traffic jams — picks free roads not short roads
Distance Vector	Rumour spreading — tell neighbors, they update, slow convergence
Broadcast	School PA system — everyone hears. Use Spanning Tree = no echo!
Multicast	WhatsApp group — only members get the message. Prune the rest!
IPv4 vs IPv6	32-bit old PIN vs 128-bit new code. 4.3B vs 340 Undecillion!